Chapter 1
QR Code Standard

1.1 Front Matter

1.1.1 Foreword

This chapter is an independent compilation and reinterpretation of the QR Code standard. It is informed by publicly available resources such as research papers, technical books, developer documentation, and collaborative knowledge bases. While I did not have access to the complete ISO/IEC publications due to their cost and limited availability, this document draws upon secondary sources and open references to provide an accessible explanation of QR Code structure and encoding principles.

Where appropriate, annotations and commentary have been added to clarify complex topics and to make the material approachable for readers who may wish to implement their own encoder or decoder, or who are simply curious about how QR codes function.

This is not an official or authoritative standard. It should be read as a study guide and commentary rather than as a normative reference.

Happy scanning, bestie **:3**

1.1.2 Disclaimer

This chapter is an independent, non-official reinterpretation of the QR Code standard, intended for educational and reference purposes. It is not affiliated with, endorsed by, or a substitute for any publication by the International Organization for Standardization (ISO), the International Electrotechnical Commission (IEC), or any related standards body.

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1.1.3 Introduction

According to **wikipedia-qr**, a QR Code (Quick Response Code) is a two-dimensional matrix barcode invented in 1994 by Masahiro Hara of the Japanese company Denso Wave, originally for tracking automobile parts. QR Codes are recognizable by their square modules arranged on a contrasting background, typically with position-detecting fiducial markers in three corners.

Data encoded within a QR Code is interpreted by imaging devices such as cameras, and recovered using error—correction algorithms—most notably Reed–Solomon error correction—to ensure robustness against noise and distortion. The encoded information is embedded in both the horizontal and vertical dimensions of the symbol, enabling compact and efficient data storage and retrieval.

1.2 Definition of Terms

1.2.1 Abbreviations

2D	two-dimensional (**mw-2d**)
QR code	quick-response code (**wikipedia-qr**)

API Application Programming Interface (**wikipedia-api**)

1.3 Symbol Description

QR code is a 2D barcode with the following characteristics:

1. Formats:
   a. QR code, the full version; and
   b. Micro QR code, a smaller version of the QR code standard for applications where symbol size is limited.
2. Character Set(**wikipedia-qr**):
   a. Numeric: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
   b. Alphanumeric: 0–9, A–Z (upper-case only), space, $ (dollar sign), % (percent symbol), * (asterisk), + (plus symbol), - (dash), . (period), / (slash), : (colon)
   c. Binary/byte: **??** as described in Appendix **??**.
   d. Kanji/kana: **??** as described in Appendix **??**.

1.4 Modes

1.5 Character Sets

1.5.1 ISO/IEC 8859-1

According to **wikipedia-ISO-8859-1**, ISO/IEC 8859-1:1998, *Information technology—8-bit single-byte coded graphic character sets—Part 1: Latin alphabet No. 1*, is part of the ISO/IEC 8859 series of ASCII-based standard character encodings, first edition published in 1987. ISO/IEC 8859-1 encodes what it refers to as "Latin alphabet no. 1", consisting of 191 characters from the Latin script.

1.5.1.1 Decode (Code → Character)

1. Start with the hex code. Example: **`0x2F`**
2. Remove the prefix **`0x`**.
3. Split into two nibbles:
   a. High nibble = first hex digit (**`2`**).
   b. Low nibble = second hex digit (**`F`**).
4. Convert the high nibble into the row by multiplying it by 16 (hex shift). Example: **`2`** → **`20`**
5. The low nibble is the column. Example: **`F`**.
6. Locate the character at row **`20`**, column **`F`** in the table.

1.6 Versions

1.6.1 Explanation

According to Denso Wave Incorporated, QR Code versions determine the data capacity and structure of a code**denso-wave-qr-versions**.

The following tables are sourced from the official QR Code documentation provided by Denso Wave.

The figures in the Numeral, Alphanumeric, Binary and Kanji columns indicate the maximum allowable number of respective characters, including those for the data bit number.

For example, when using the Version 1 QR Code with correction level L, the maximum allowable characters for data bit number, numerals, binaries, and Kanji are 152, 25, 17, and 10 respectively.

As explained by Denso Wave in the aforementioned documentation, to determine the version of the QR code to use, suppose the data to be input consists of 100-digit numerals. This can be achieved by following the steps described below**denso-wave-qr-versions**:

1. In the QR Code table, choose "Numeric" as the type of input data.
2. Choose an error correction level from the options of L, M, Q and H. For example, the error correction level that will be used is M.
3. Find a figure in the table, 100 or over and the closest to 100 that is at the intersection with a correction level M row. The number of the version row that contains this figure is the most appropriate version number. We can see at the tables below that the version closest to that is Version 3 with capacity for up to 101 numeric characters at error correction M.